

Vedanth Dharnish

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EDUCATION

Texas A&M University

College Station, TX

*B.S. Computer Science, **Craig and Galen Brown Engineering Honors***

May 2028

Coursework: Data Structures & Algorithms, Analysis of Algorithms, Computer Systems, Computer Architecture, Software Engineering, Functional Programming, Machine Learning and Predictive Modeling

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL, Haskell, Assembly, MATLAB, HTML/CSS

ML/AI: PyTorch, TensorFlow, scikit-learn, BERT, Whisper, Silero VAD, YOLOv8, OpenCV, NumPy, Pandas, NLP

LLMs & Agents: Ollama, MCP, LangChain, tool/function calling, RAG, prompt engineering, schema validation

Systems & Backend: ROS 2, WebSockets, REST/JSON APIs, real-time audio streaming (PCM16),

asyncio/concurrency, Flask, Node.js, React, Firebase, Unity

Tools: Git/GitHub, Docker, Linux, pytest, unit & integration testing, VS Code, Jupyter

EXPERIENCE

AI Engineering Intern

Jun. 2026 – Aug. 2026

Cacti Industries

Austin, TX

- Shipped the voice-command decision layer for an XR-controlled robot fleet, classifying operator speech into **15** schema-validated skills via a **local LLM (Ollama)**; now in production.
- Eliminated silent command loss—merged utterances dropping safety-critical commands like “cancel”—with a streaming **Silero VAD** pipeline over **WebSockets**, tuned on a **25-clip** noise corpus.
- Decoupled skills from the LLM host with a **Model Context Protocol** server auto-registering **16** tools at runtime.
- Cut spurious replies with follow-up state tracking and fragment filtering, validated by a **56-test pytest** suite.

Undergraduate Teaching Assistant

Aug. 2026 – Present

Texas A&M University

College Station, TX

- Teach **Python** fundamentals to **100+** first-year engineering students through weekly labs and office hours.
- Evaluate programming assignments and give written feedback on **software design**, code quality, and debugging.

Undergraduate Research Assistant — Phishing Detection

May 2026 – Present

Texas A&M University

College Station, TX

- Shipped 2 feature extractors (**BERT** persuasion, **transformer** emotion) into a multi-modal phishing framework.
- Raised detection **F1** 0.599 → 0.756 by fusing 26 persuasion/emotion features into **scikit-learn** ensembles, 5-fold CV.
- Benchmarked 6 feature pipelines, showing sentiment adds no lift over bag-of-words (0.926 → 0.919).

Undergraduate Research Assistant — Model Generalization

Mar. 2025 – May 2025

Texas A&M University

College Station, TX

- Analyzed generalization vs. memorization across **100+ neural network architectures** and model families.
- Built **Python** visualization and colormap pipelines, improving color-mapping accuracy to **95%**.
- Automated **matplotlib** workflows and data-validation checks, cutting report-generation time by **60%**.

PROJECTS

Deterministic Replay Debugger for LLM Agents | Python, httpx, MCP, LangChain, pytest

Aug. 2026

- Published an open-source agent debugger to **PyPI (v1.0, 8 releases)** replaying failed runs offline at **60×** speed.
- Intercepted nondeterminism at the **httpx/urllib3** transport layer, covering OpenAI, Anthropic, and **Bedrock** without SDK patching.
- Recorded **MCP** and **LangGraph** sessions as first-class events, enabling tool-set diffs and replay.
- Shipped **882 tests** at **95%** coverage across Python 3.9–3.13, gated by a symlinked-path wheel install in CI.

Traffic Incident Monitoring Dashboard | Python, OpenCV, Flask, React, hls.js, Vercel

Oct. 2025

- Deployed a dashboard streaming **12 live** DOT camera feeds across **4 state agencies** via **hls.js**.
- Derived incident windows with **OpenCV** background modeling and centroid tracking, validated frame-by-frame.
- Built pluggable alerting (**Discord**, **Twilio**) with ordered fallback, on a single **Vercel** serverless deploy.

LEADERSHIP

Engineering Honors Executive Committee — Professional Development

Jan. 2026 – Present

Texas A&M University

College Station, TX

- Co-organize the EH Career Fair and Research Lab Open House for the Engineering Honors student body.
- Coordinate volunteers and event logistics to support professional development programming for students.